

Apriori: Improvements and Alternatives

❑ Reduce passes of transaction database scans

- ❑ **Partitioning** (e.g., Savasere, et al., 1995)

- ❑ Dynamic itemset counting (Brin, et al., 1997)



To be discussed in subsequent slides

❑ Shrink the number of candidates

- ❑ **Hashing** (e.g., DHP: Park, et al., 1995)

- ❑ Pruning by support lower bounding (e.g., Bayardo 1998)

- ❑ Sampling (e.g., Toivonen, 1996)



To be discussed in subsequent slides

❑ Exploring special data structures

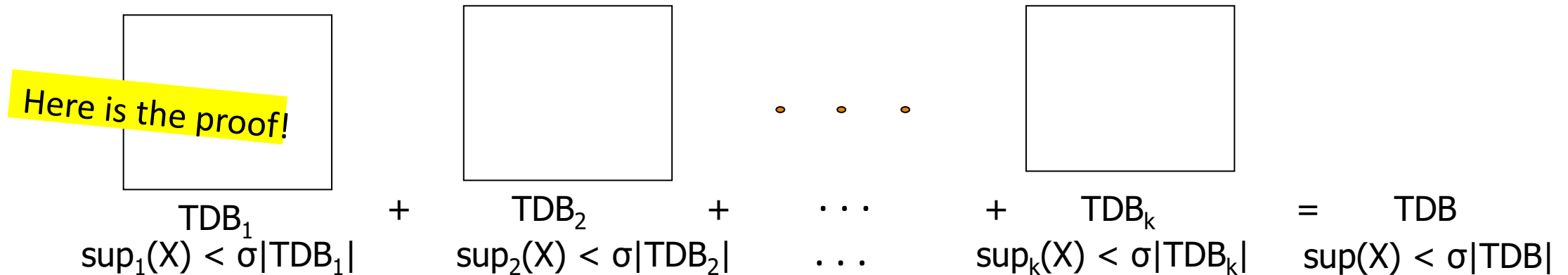
- ❑ Tree projection (Agarwal, et al., 2001)

- ❑ H-miner (Pei, et al., 2001)

- ❑ Hypercube decomposition (e.g., LCM: Uno, et al., 2004)

Partitioning: Scan Database Only Twice

- Theorem: *Any itemset that is potentially frequent in TDB must be frequent in at least one of the partitions of TDB*



- Method: Scan DB twice (A. Savasere, E. Omiecinski and S. Navathe, *VLDB'95*)
 - Scan 1: Partition database so that each partition can fit in main memory (why?)
 - Mine local frequent patterns in this partition
 - Scan 2: Consolidate global frequent patterns
 - Find global frequent itemset candidates (those frequent in at least one partition)
 - Find the true frequency of those candidates, by scanning TDB_i one more time

Direct Hashing and Pruning (DHP)

- ❑ **Hashing:** $v = \text{hash}(\text{itemset})$
- ❑ **1st scan:** When counting the 1-itemset, hash 2-itemset to calculate the bucket count
- ❑ **Example:** At the 1st scan of TDB, count 1-itemset, and hash 2-itemsets in the transaction to its bucket

- ❑ {ab, ad, ce}
- ❑ {bd, be, de}
- ❑ ...

- ❑ At the end of the first scan,

- ❑ if $\text{minsup} = 80$, remove *ab, ad, ce*, since $\text{count}\{ab, ad, ce\} < 80$

V might be same for different itemset

Itemsets	Count
{ab, ad, ce}	35
{bd, be, de}	298
.....	...
{yz, qs, wt}	58

Hash Table



Check the minsup

A key observation: A k -itemset cannot be frequent if its corresponding hashing bucket count is below the *minsup* threshold